This is most often referred to as the *Itō formula*. In recognition of the earlier contributions of Doeblin, many authors now add the second name, and we will follow this convention, referring to (6.17) as the *Itō-Doeblin formula*.[6]

We also mention that the Itō-Doeblin formula can be easily generalized to functions of the form $\phi(x, t)$:

$$\mathrm{d}\phi(\mathrm{X}, t) = \phi_t(\mathrm{X}, t) + \phi_x(\mathrm{X}, t)(a(\mathrm{X}, t)\mathrm{d}t + b(\mathrm{X}, t)\mathrm{dW}) + \tfrac{1}{2}\phi_{xx}(\mathrm{X}, t)b(\mathrm{X}, t)^2\mathrm{d}t. \tag{6.18}$$

6.3.2 Distributional Solution of the Ornstein-Uhlenbeck SDE

For now we will examine the special case of a scalar equation

$$\mathrm{dX} = -\gamma \mathrm{X}\mathrm{d}t + \sigma \mathrm{dW}, \tag{6.19}$$

where $\gamma > 0$ and σ are constants. We term Eq. (6.19) the Ornstein-Uhlenbeck SDE.

We saw before that we could think of the simple SDE initial value problem (6.8) as having a solution defined by a certain random process. In the same way we would like to obtain, for the Ornstein-Uhlenbeck SDE (6.19), an explicit stochastic process which is in some sense equivalent to solving the SDE. Multiply both sides of (6.19) by the integrating factor $\exp(\gamma t)$, and observe that

$$\mathrm{d}\left(\exp(\gamma t)\mathrm{X}\right) = \exp(\gamma t)(\mathrm{dX} + \gamma \mathrm{X}\mathrm{d}t),$$

thus (6.19) becomes

$$\mathrm{d}\left(\exp(\gamma t)\mathrm{X}\right) = \sigma \exp(\gamma t)\mathrm{dW}(t).$$

We then integrate both sides to obtain

$$\exp(\gamma t)\mathrm{X}(t) = \mathrm{X}(0) + \int_0^t \sigma \exp(\gamma s)\mathrm{dW}(s).$$

Multiplying by $e^{-\gamma t}$ gives

$$\mathrm{X}(t) = e^{-\gamma t}\mathrm{X}(0) + \sigma e^{-\gamma t}\int_0^t \exp(\gamma s)\mathrm{dW}(s).$$

[6] A convenient method for remembering it is to expand $\mathrm{d}\phi(\mathrm{X}) \stackrel{\mathrm{def}}{=} \phi(\mathrm{X} + \mathrm{dX}) - \phi(\mathrm{X})$ in powers of dX up to the second order, then substitute for dX from the SDE (6.14) and finally set all higher powers and products of dt and dW to zero with the one exception $\mathrm{dW}^2 = \mathrm{d}t$.